



**Java OOPs:
Calculator App Using OOPs
Assignment - 3**

Problem Statement:

You have been assigned to develop a robust Object-Oriented Programming (OOP) based Basic Calculator application in Java. The goal of this project is to create a user-friendly calculator that can perform standard arithmetic operations, including addition, subtraction, multiplication, and division.

Requirements:

1. Create a Calculator class with methods for each arithmetic operation:
 - a. **add**: Accepts two numbers and returns their sum.
 - b. **subtract**: Accepts two numbers and returns the result of subtracting the second number from the first.
 - c. **multiply**: Accepts two numbers and returns their product.
 - d. **divide**: Accepts two numbers and returns the result of dividing the first number by the second. Ensure to handle division by zero gracefully.
2. Implement a user interface in a separate Main class for user interactions. The user interface should:
 - a. Display a menu with options for each arithmetic operation.
 - b. Prompt the user to choose an operation and enter two numbers as operands.
 - c. Call the corresponding method in the Calculator class to perform the operation.
 - d. Display the result to the user.
 - e. Handle invalid user inputs gracefully and provide informative error messages.
 - f. Ensure that the calculator provides accurate results for all arithmetic operations, preserving the correct number of decimal places.
 - g. Create a well-documented code with appropriate comments, adhering to Java coding conventions and best practices.